

Skytree GUI Quick Start

Release 15.4.0

SKYTREE®

Notices

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Customers with a support contact can contact support using any of the following methods:

- Log in to support.skytree.net.
- Send questions to support@skytree.net.
- Call 408.392.9300 and select option 3.
- You can also view additional Skytree documentation by logging on to support.skytree.net.

Freemium customers can send inquiries to the LinkedIn® Community Group for Skytree:

<https://www.linkedin.com/groups/4468542>

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Getting Started with the GUI

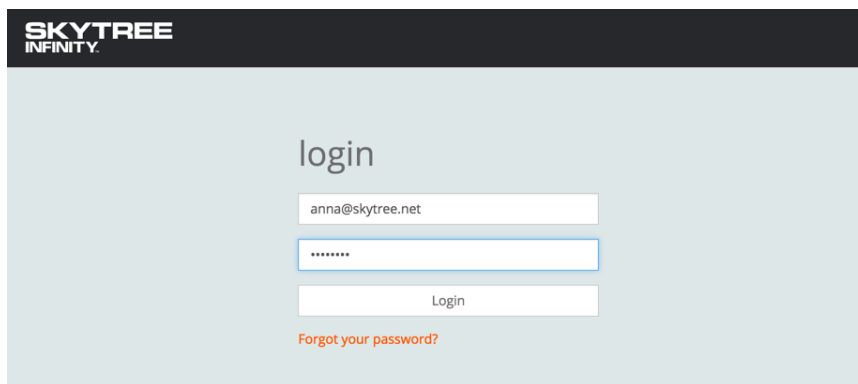
This document describes how to get started using the Skytree GUI. For detailed information about all of the functions available in the GUI, refer to the *Skytree GUI Guide*, which is available via the HELP menu item in the GUI.

Using the included income.data file, this document will take you through the following steps:

1. Logging in
2. Creating a project
3. Adding the income.data dataset to the project
4. Transforming the dataset
5. Training a regression model using AutoModel
6. Testing the model
7. Viewing results

Step 1. Logging In

1. In a supported browser, navigate to Skytree on your server.
2. Log in using the e-mail and password that were provided by your Admin.

A screenshot of the Skytree login interface. At the top, there is a dark header with the 'SKYTREE INFINITY' logo. Below the header, the word 'login' is centered. There are two input fields: the first contains the email 'anna@skytree.net' and the second is a password field with masked characters. Below the password field is a 'Login' button. At the bottom of the login area, there is a link that says 'Forgot your password?' in orange text.

Step 2. Creating a Project

1. After you successfully log in, the Skytree GUI opens to the Projects page. To create a new project, click the **+ NEW PROJECT** button.
2. Enter a unique name and a description for the project and then select **Create**. Note that you will receive an error if you attempt to specify a Project Name that already exists.

New Project

Project Name

Description

Cancel

Create

Upon successful completion, the new Project Card will appear on the Projects page. At this point, you can begin adding datasets and training models.

Projects

+ New Project

Search

PREDICTION

Project for predictions

Created - 2015-12-03 02:39 PM

Last Modified - 2015-12-03 02:39 PM

DATASETS	MODELS	TEST RESULTS
0	0	0

LEAD SCORING - ADVANCED

Find me more new customers

Created - 2015-12-01 10:40 AM

Last Modified - 2015-12-02 10:45 AM

DATASETS	MODELS	TEST RESULTS
7	7	1

LEAD SCORING - BASIC

Find new customers

Created - 2015-12-01 10:22 AM

Last Modified - 2015-12-01 10:44 AM

DATASETS	MODELS	TEST RESULTS
3	3	0

LEAD SCORING

Find me new customers

Created - 2015-11-30 01:04 PM

Last Modified - 2015-12-01 10:21 AM

DATASETS	MODELS	TEST RESULTS
6	3	0

FRAUD DETECTION

Point of sale credit card swipe fraud

Created - 2015-11-30 12:14 PM

Last Modified - 2015-11-30 01:01 PM

DATASETS	MODELS	TEST RESULTS
5	4	0

Step 3. Adding a Dataset to Your Project

The income.data file is included with Skytree Platform. This file can be found under `/examples`. Perform the following steps to import the income.data dataset.

1. Click anywhere in the Project Card that was just created. This opens a page listing all the datasets, models, and results that were added to the project (initially empty).

SKYTREE

PREDICTION

Projects

Prediction

Datasets

PREDICTION

Project for predictions

DATASETS

MODELS

RESULTS

PLOTS

+ New Dataset

Search

2. Click the **+ NEW DATASET** button to add a new dataset.

Projects > Prediction > Datasets > New Dataset

Location: Register HDFS

Name:

HDFS:

☐ First Line Has Column Names

Separator: Comma

☐ Has Missing Value Identifier

[Preview](#)

3. Select **Import Local File** from the dropdown menu on the left.
4. Click the **Select file** button.
5. Browse to the file location where Skytree was installed, and navigate to the `/examples` folder. Locate the `income.data` file, then click **Choose**. Once selected, the Name field defaults to the file name. This field is editable.
6. The `income.data` file includes column names. Values in this file are separated by commas, and missing values are specified using `"?"`.

Location: Import Local File

File: [Clear Selection](#)

Raw Data View

```
age, workclass, fnlwgt, education, education-num, marital-status, occupation, relati
39, State-gov, 77516, Bachelors, 13, Never-married, Adm-clerical, Not-in-family, Whi
50, Self-emp-not-inc, 83311, Bachelors, 13, Married-civ-spouse, Exec-managerial, Hus
38, Private, 215646, HS-grad, 9, Divorced, Handlers-cleaners, Not-in-family, White,
53, Private, 234721, 11th, 7, Married-civ-spouse, Handlers-cleaners, Husband, Black,
28, Private, 338409, Bachelors, 13, Married-civ-spouse, Prof-specialty, Wife, Black,
37, Private, 284582, Masters, 14, Married-civ-spouse, Exec-managerial, Wife, White,
49, Private, 160187, 9th, 5, Married-spouse-absent, Other-service, Not-in-family, B
```

Name:

☒ First Line Has Column Names

Separator: Comma

☒ Has Missing Value Identifier

Missing Value Identifier: [Preview](#)

7. Click **Preview**. A preview of the data displays showing you a tabular format of up to the first 100 lines of data.
8. Click **Import** to add that dataset to the project. The Datasets page will display indicating the status of the import (right-most column). The status will initially show INPROGRESS and will change to READY when the import completes successfully.

MODELS	STATUS
0	INPROGRESS : Sample . View Job Status
0	READY

Step 4. Transform the Dataset

The next step is to transform the dataset by adding a unique ID column to the income.data file. An ID column is required when performing Predict & Evaluate operations.

1. From the list of datasets in your project, hover over the dataset you just added and select **Transform**.

Filter Columns: Generate ID Column

NAME	UNIQUE ID	CATEGORICAL	BLOB	REMOVE
ID	☒	☐	☐	☐
age	☐	☐	☐	☐
workclass	☐	☒	☐	☐
fnlwgt	☐	☐	☐	☐
education	☐	☒	☐	☐
education-num	☐	☐	☐	☐

Preview

```
{
  "transform": {
    "inputs": [
      "4274348784894814824"
    ],
    "transformSteps": [
      {
        "columnName": "ID",
        "type": "AddUniqueIdColumn"
      }
    ]
  },
  "sourceId": null,
  "projectId": "4528433328984416731",
  "names": [
    "income.data.ids"
  ]
}
```

New Dataset Name:

Cancel Transform Dataset

2. Click the **Generate ID Column** button.
3. Enter a name for the column (for example, "ID").
4. Optionally specify a new name for this dataset (for example, "income.data.ids").
5. Click **Transform Dataset** when you are done.

Step 5. Adding a Model

You create a model by selecting the Objective Column, which is the field you want to predict, as well as additional configuration options to use for tuning and training.

During the model creation process, the Skytree GUI automatically determines whether the problem you are trying to solve is classification or regression. This is based on whether the Objective Column is numeric or categorical.

Perform the following steps to add a model to a project:

1. From the Models page, click the **+ NEW MODEL** button.

Note: You can also create a new model when viewing a project's list of datasets. Select a dataset, then click **Create model** to start the model creation process.

2. Select the income.data dataset.
3. Specify a descriptive name for the model, such as Income Prediction.
4. Select the Objective Column. This represents the column to be predicted. Based on the selected Objective Column, the Skytree GUI automatically determines whether the problem to be solved is a classification or regression problem. In this example, we are selecting the "yearly-income" column, thus ensuring a classification problem.
5. Select the machine learning method to be used to build the mode. Available machine learning methods for classification problems include:

AutoModel, GBT, RDF, SVM, and GLM

For this example, we will use AutoModel and the default values.

Note: Refer to the Command Reference Appendix in the *Skytree GUI Guide* for information about the options available for each machine learning method. If an incorrect set of parameters is selected, a message will appear beside the option, and an  icon on the related configuration sections.

6. Click **Learn** to build the model.

Dataset	income.data	Select										
Model Name	Income Prediction											
Objective Column	yearly-income ▾											
ML Task	Classification											
Method	Automodel ▾											
Search Iterations	100											
Restrict tuning parameter space	☒											
Model validation type	Holdout ▾											
Holdout ratio	0.2											
Classifier testing objective	Gini ▾											
Classification objective	F-Score ▾											
k for Precision@k	0.1											
Reproducibility seeds	Show more ▾											
Enable PMML	☒											
Yarn Configuration	<table border="1"> <tr> <td>Number of nodes</td> <td>3</td> </tr> <tr> <td>Available memory for each node</td> <td>50 GB</td> </tr> <tr> <td>Available virtual cores for each node</td> <td>16</td> </tr> <tr> <td>Maximum memory for each container</td> <td>50 GB</td> </tr> <tr> <td>Maximum virtual cores for each container</td> <td>32</td> </tr> </table>		Number of nodes	3	Available memory for each node	50 GB	Available virtual cores for each node	16	Maximum memory for each container	50 GB	Maximum virtual cores for each container	32
Number of nodes	3											
Available memory for each node	50 GB											
Available virtual cores for each node	16											
Maximum memory for each container	50 GB											
Maximum virtual cores for each container	32											
Estimated Resource Requirements	<table border="1"> <tr> <td>Number of containers</td> <td>1</td> </tr> <tr> <td>Memory for each container</td> <td>50 GB</td> </tr> <tr> <td>Virtual cores for each container</td> <td>16</td> </tr> </table>		Number of containers	1	Memory for each container	50 GB	Virtual cores for each container	16				
Number of containers	1											
Memory for each container	50 GB											
Virtual cores for each container	16											
Learn												

The new model will appear immediately on the Models list. The Status column will show the model's progress while it is building and will change to READY when the new model has been built.

Step 6. Predict & Evaluate

Perform the following steps to generate a new prediction file:

1. From the Results page, click the **+ New Predict & Evaluate** button.
2. Select the Model to be used as the basis for the prediction.
3. Enter a name for the generated results.
4. Select the Dataset that will be compared against the model. This should be the dataset that you transformed with a unique ID column.

Note: Only datasets that with an ID column can be specified.

5. Click **Predict & Evaluate** when you are ready.

Projects > Prediction > Results > **New Predict & Evaluate**

Model: Income Prediction Select

Method: Automodel (GBT)

Objective column: yearly-income

Result name: Yearly Income Prediction

Dataset: income.data.ids Select

ID column: id

Environment: Default (1 x [4098 MB, 1 core]) Select

Predict & Evaluate

The new result will appear immediately on the Results list. The Status column shows the progress of the result as it is building. This will change to READY when the new result is available, and the Test Scores column will populate with the results.

DATASETS MODELS RESULTS PLOTS + New Predict & Evaluate Search								
LAST UPDATED	NAME	ID	MODEL ID	TEST DATASET ID	OBJECTIVE COLUMN	METHOD	STATUS	TEST SCORES
2015-12-03 03:21 PM	Yearly Income Prediction	63214772729728361...	41066280399914192...	675207061566701409	yearly-income	Automodel (GBT)	READY	Gini: 0.8890 Accuracy: 0.8877

Step 7. Viewing Results

READY test results will appear in a table on the Results page. This table shows the Model and Test Dataset IDs that were used during the test, the target column, the machine learning method, and the test scores. Click on a column heading to sort the data in that column. Mouse over your new test result in this list to view the following additional options:

- **View details:** View additional information about the test, including a results summary, ROC curve, confusion matrix, and scores. Note that the information that displays can vary depending on the machine learning method used to produce the result. Similarly, if you performed a Predict only, then scoring information (including confusion matrix and ROC curve) will not be available.
- **View graph:** View a graphical representation of this test result's history, from importing the original dataset through testing.
- **View logs:** Review the log files that were created when the result was generated. A log is available when the result is no longer INPROGRESS.
- **Delete:** Delete the result (including all associated files).

DATASETS MODELS RESULTS PLOTS + New Predict & Evaluate Search								
LAST UPDATED	NAME	ID	MODEL ID	TEST DATASET ID	OBJECTIVE COLUMN	METHOD	STATUS	TEST SCORES
2015-12-03 03:21 PM	Yearly Income Prediction	View details View graph View logs Delete...	1066280399914192...	675207061566701409	yearly-income	Automodel (GBT)	READY	Gini: 0.8890 Accuracy: 0.8877

What's Next?

Now that you have finished your first Skytree GUI example, you are ready to begin creating models using your own data. Refer to the *Skytree GUI Guide* for information on all of the features, model building methods, and configuration options available in Skytree. This guide is available via the GUI Help menu.